A COUNTEREXAMPLE TO GENERIC INJECTIVITY OF THE TANGENT GAUSS MAP

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We work over $\mathbb{C}$. For a curve $C \subset A$ in an abelian variety, the tangent Gauss map sends a smooth point to its tangent line, translated to the origin:

$$\phi_C : C^{\mathrm{sm}} \longrightarrow \mathbb{P}(T_0 A), \qquad p \longmapsto d(t_{-p})_p(T_p C).$$

We call C nondegenerate if it is not contained in a translate of a proper abelian subvariety.

Theorem 1. *There exists a nondegenerate curve in an abelian threefold whose tangent Gauss map has generic degree* 3.

1. Strategy

Let C be a smooth nonhyperelliptic curve of genus 4, and let $\mathcal{L}$ be a line bundle of degree 3 with $h^0(\mathcal{L}) = 2$. It is base-point-free, since a base point would give a degree-two pencil. Thus $|\mathcal{L}| : C \longrightarrow \mathbb{P}^1$ has degree 3. We seek the following two properties:

$$\mathcal{L}^{\otimes 2} \cong \omega_C, \qquad \operatorname{Sym}^2 \mathrm{H}^0(\mathcal{L}) = q^* \mathrm{H}^0(\Omega^1_A) \quad \text{for a quotient } q : \operatorname{Jac}(C) \longrightarrow A.$$

Here we identify $\operatorname{Sym}^2 \mathrm{H}^0(\mathcal{L})$ with its image in $\mathrm{H}^0(\omega_C)$ under multiplication. Its dimension is 3: a relation between the three products of a basis would force the ratio of its sections to be constant. We call the second property the *cotangent-space condition*. The resulting linear system defines

$$\phi : C \xrightarrow{|\mathcal{L}|} \mathbb{P}^1 \xrightarrow{\nu_2} \mathbb{P}^2,$$

where ν_2 is the quadratic Veronese embedding. We will realize ϕ as a tangent Gauss map.

Start with a smooth hyperelliptic curve C_0 of genus 4 admitting a tangential cover $\pi_0 : (C_0, p_0) \longrightarrow (E_0, 0)$ at a Weierstrass point. Such covers exist, for example among the Lamé spectral curves [3, Table 3 and Theorem L] and [2, Proposition 3.1]. Tangentiality means that $\pi_0^* E_0 \subset \operatorname{Jac}(C_0)$ is tangent to the Abel–Jacobi image at p_0. Put $\mathcal{L}_0 = \mathcal{O}_{C_0}(3p_0)$ and $A_0 = \operatorname{Jac}(C_0)/\pi_0^* E_0$, with quotient map q_0. Then

$$\mathcal{L}_0^{\otimes 2} \cong \omega_{C_0}, \qquad \operatorname{Sym}^2 \mathrm{H}^0(\mathcal{L}_0) = \mathrm{H}^0\left(\omega_{C_0}(-2p_0)\right) = q_0^* \mathrm{H}^0(\Omega^1_{A_0}).$$

Indeed, the products of $1, x \in \mathrm{H}^0(\mathcal{L}_0)$ span $\mathrm{H}^0\left(\mathcal{O}_{C_0}(4p_0)\right)$, where x is a hyperelliptic coordinate with a double pole at p_0. Also, $\mathrm{H}^0\left(\omega_{C_0}(-2p_0)\right) = \mathrm{H}^0\left(\omega_{C_0}(-p_0)\right)$ is the annihilator of the tangent line at p_0. Conversely, if this annihilator comes from an abelian quotient, its connected kernel is an elliptic curve tangent to C_0; dualizing the inclusion and composing with the Abel–Jacobi map gives a tangential cover.

2. A dimension count for deformations

Let B be a small simply connected marked deformation space of the spin curve $(C_0, \mathcal{L}_0)$. Its dimension is $3g - 3 = 9$. The theta-null locus

$$S = \left\{ t \in B \;\middle|\; h^0(\mathcal{L}_t) \geqslant 2 \right\}$$

is locally defined by one theta constant. After shrinking B, upper semicontinuity gives $h^0(\mathcal{L}_t) = 2$ on S. Put

$$W_t := \operatorname{Sym}^2 \mathrm{H}^0(\mathcal{L}_t) \subset \mathrm{H}^0(\omega_{C_t});$$

these spaces form a holomorphic subbundle of rank 3.

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Set

$$V_0 = q_0^* \,\mathrm{H}^1(A_0;\mathbb{Q}) \subset \mathrm{H}^1(C_0;\mathbb{Q}),$$

and let $V_t \subset \mathrm{H}^1(C_t;\mathbb{Q})$ be its parallel transport under the Gauss–Manin connection. To preserve the cotangent-space condition, we use the following equivalent formulations:

$$\begin{aligned}
&\begin{cases} V_t \text{ remains a Hodge substructure,} \\ V_{t,\mathbb{C}} \cap \mathrm{H}^0(\omega_{C_t}) = W_t \end{cases} \\
\iff & V_{t,\mathbb{C}} = W_t \oplus \overline{W}_t \\
\iff & W_t \subseteq V_{t,\mathbb{C}} \\
\iff & \mathrm{rk}\left[W_t \longrightarrow \mathrm{H}^1(C_t;\mathbb{C})/V_{t,\mathbb{C}}\right] = 0.
\end{aligned}$$

The middle equivalence follows from rationality of V_t and Hodge type, since $\dim W_t = 3$ and $\dim V_t = 6$. The last condition is the vanishing of a bundle map of ranks 3 and 2, so it imposes at most six equations on S. Consequently, every irreducible component Z of its zero locus through the seed satisfies

$$\dim Z \geqslant 9 - 1 - 6 = 2.$$

On Z, the Hodge substructure V_t gives an abelian quotient $q_t : \mathrm{Jac}(C_t) \longrightarrow A_t$ with connected kernel and $q_t^* \,\mathrm{H}^0(\Omega^1_{A_t}) = W_t$.

We compare this with the hyperelliptic part. On a hyperelliptic curve C_t of genus 4, a degree-three pencil must have a base point: otherwise, together with the hyperelliptic pencil it would give a birational map to a curve of bidegree $(2,3)$ in $\mathbb{P}^1 \times \mathbb{P}^1$, of arithmetic genus 2. Writing $\mathcal{L}_t = H_t(p_t)$, where H_t is the hyperelliptic pencil, the spin condition and $\omega_{C_t} \cong H_t^{\otimes 3}$ give $\mathcal{O}_{C_t}(2p_t) \cong H_t$. Hence p_t is a Weierstrass point, and

$$\mathcal{L}_t \cong \mathcal{O}_{C_t}(3p_t), \qquad W_t = \mathrm{Sym}^2 \,\mathrm{H}^0(\mathcal{L}_t) = \mathrm{H}^0\left(\omega_{C_t}(-2p_t)\right).$$

By the tangency criterion above, these points give tangential covers $\pi_t : (C_t, p_t) \longrightarrow (E_t, 0)$. Their degree is locally fixed: the integral homology lattice of the elliptic kernel of q_t is the annihilator of V_t, fixed under the marking, and dualizing its inclusion fixes the cohomology map of π_t and its cup-product pairing.

For a fixed elliptic curve and degree there are only finitely many hyperelliptic tangential covers [4, Proposition 2.5, 3.; Section 2, Remark 1, 3.–4.]. Indeed, the relation $2n + 1 = \sum_{i=0}^{3} \mu_i^2 + 4d$ leaves finitely many types for a fixed degree n, and each corresponding Severi set is finite. Since elliptic curves have one modulus and the choices of p_t are finite,

$$\dim\left(Z \cap \mathcal{H}_4\right) \leqslant 1 < 2 \leqslant \dim Z.$$

We can therefore choose a nonhyperelliptic point of Z.

For this point, compose the Abel–Jacobi map based at $b \in C$ with q: $f = q \circ j_b : C \longrightarrow A$. The pulled-back invariant forms are the products of sections of $\mathcal{L}$, so f is immersive and its tangent map is ϕ. Its image contains the origin and generates A, hence is nondegenerate. If f had degree $e > 1$ onto the normalization of its image, then $e \mid 3$, since ϕ factors through that normalization. Thus $e = 3$, and the normalization would map with degree 1 to the conic $\nu_2(\mathbb{P}^1)$. This is impossible in an abelian variety. Therefore C is the normalization of $f(C)$ and the tangent Gauss map of $f(C)$ has generic degree 3, proving Theorem 1.

3. Simplicity and symmetries

Theorem 2. *The abelian threefold can be chosen simple. The resulting curve is preserved by no nonidentity automorphism of A with scalar differential.*

First choose the hyperelliptic seed so that A_0 is simple; the existence of such a seed is checked below. For every nonzero proper rational subspace $R \subset V_0$ on which the polarization is nondegenerate, let e_R be the orthogonal projector. The condition

$$e_R(W_t) \subset W_t$$

defines a closed analytic subset of Z. It is proper because it misses the seed, and it holds precisely when R is a Hodge substructure. Every abelian subvariety of A_t gives such a substructure. There are only countably many R, so Baire's theorem allows us to choose a point outside these subsets and $\mathcal{H}_4$. Its quotient A is simple.

Now suppose $\tau \in \mathrm{Aut}(A)$ preserves $f(C)$ and has scalar differential, using translations to identify tangent spaces. It lifts to an automorphism σ of C with $\phi \circ \sigma = \phi$. If σ is the identity, then τ fixes a generating curve pointwise and is the identity. Otherwise, the order of σ divides $\deg \phi = 3$, so it is 3 and $C/\langle\sigma\rangle \cong \mathbb{P}^1$. Riemann–Hurwitz gives a fixed point of σ. Taking its image as the origin makes τ a group automorphism u of order 3. Its scalar differential is a primitive cube root of unity λ. Then $\mathbb{Q}(\lambda)$ acts on $\mathrm{H}_1(A;\mathbb{Q})$, and the complex structure on real homology is $(u_* - \mathrm{Re}(\lambda)) \,/\, \mathrm{Im}(\lambda)$. Every one-dimensional $\mathbb{Q}(\lambda)$-subspace therefore has a complex-invariant realification and defines an elliptic subvariety of A, contradicting simplicity. This proves the symmetry assertion.

Existence of a seed with simple quotient. Take $E_0 : y^2 = 4x^3 - 4x - 1$, let e_1, e_2, e_3 be the roots of $4e^3 - 4e - 1$, and take the smooth projective model of

$$C_0 : \quad \nu^2 = (B^3 - 208B + 560) \prod_{i=1}^{3} \left(B^2 + 10e_i B - 35e_i^2 - 28\right).$$

This is the genus-four Lamé spectral curve, with a degree-ten cover of E_0 tangential at infinity [3, Table 3 and Theorem L] and [2, Proposition 3.1 and the discussion after (17)]. The cover is defined over $\mathbb{Q}$, and E_0 has discriminant 37. Modulo 11, the polynomial on the right is the square-free polynomial $B^9 + B^4 + 6B^3 + 9B^2 + 7B + 5$; hence C_0 has genus 4 and good reduction. Direct counting gives

$$(\#C_0(\mathbb{F}_{11^r}))_{r=1}^{4} = (10, 120, 1312, 15312), \qquad \#E_0(\mathbb{F}_{11}) = 7.$$

Newton's identities and division by the elliptic factor $T^2 - 5T + 11$ give the Frobenius polynomial of $A_0 = \mathrm{Jac}(C_0)/\pi_0^* E_0$:

$$h(T) = T^6 + 3T^5 + 5T^4 - 14T^3 + 55T^2 + 363T + 1331.$$

It is irreducible modulo 3, and $11 \nmid 14$, so this reduction is simple and ordinary. An ordinary simple abelian threefold over a finite field of odd cardinality is absolutely simple unless its Frobenius polynomial has the form $T^6 + cT^3 + q^3$ [1, Proposition 3.4]. Thus this reduction is absolutely simple, and so is A_0 in characteristic zero, since a proper abelian subvariety would specialize to one after a finite extension.

References

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